

Retrospective LLM Validation Protocol

«Explaining Blood Pressure Reduction Heterogeneity in SGLT-2 Inhibitor Trials: A Meta-Analysis Stratified by Patient Characteristics»

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Methodology and Sampling Strategy

To validate the performance of the Large Language Model (LLM) hybrid screening methodology applied in the main manuscript, a stratified retrospective analysis was conducted. All imported trials and articles from Covidence were exported, with EndNote assigning a record number for each entry. These assigned record numbers for each screening phase were used to ensure an unbiased study selection by using a Random Number Generator (RNG) to ensure an objective and representative dataset. The generated list of numbers was then sorted from low to high, for simplicity in finding the correct article.

To assess the performance of the LLM independent of human oversight or assistance, the LLM was given one chance to assess an article as «Include» or «Exclude», when given all relevant documents and files for an article. If an article was incorrectly assessed, we further ask for clarification to see why it was incorrectly assessed. Regardless of whether the LLM-system retracts its original assessment or not, only the initial answer is registered. Any further probing is strictly for understanding where the system went wrong, and what could potentially work better in future iterations of the human-LLM-hybrid screening.

The human-LLM-hybrid screening validation was divided into three distinct phases to test system robustness across the entire systematic review pipeline:

1. **Title and Abstract screening (k=170):** A random sample of 10% of all non-duplicate trials (k=1704) was selected to evaluate the model's specificity. The primary objective was to assess the system's ability to correctly exclude irrelevant records within a high-volume dataset.
2. **Full-text screening (k=34):** 10% of the articles proceeding to full-text review (k=335) were validated. This stage focuses on complex exclusion criteria and evaluates model precision in "borderline" cases, potentially reducing a falsely high specificity in the previous stage, by excluding obviously irrelevant articles.
3. **Final inclusion (k=38):** All included studies underwent validation to verify the model's sensitivity. This is the most critical step to ensure no relevant studies were overlooked (false negatives).

Data availability

Publicly accessible raw data and EndNote export files for all validation phases can be found at <https://doi.org/10.5281/zenodo.18451105> for independent methodological testing.

Phase 1: Title and Abstract screening (Trial IDs: 5715-7418)

5727	5749	5750	5752	5753	5761	5762	5781	5797	5812
5829	5831	5835	5836	5837	5851	5863	5884	5885	5888
5918	5921	5925	5928	5933	5991	6001	6004	6033	6037
6062	6066	6068	6077	6088	6092	6121	6122	6138	6144
6149	6153	6159	6160	6175	6178	6185	6187	6203	6210
6213	6237	6243	6259	6269	6288	6302	6306	6314	6332
6335	6338	6341	6346	6349	6368	6369	6393	6394	6395
6396	6398	6403	6424	6428	6429	6451	6466	6467	6476
6498	6518	6526	6562	6565	6570	6607	6619	6620	6621
6627	6631	6642	6646	6651	6657	6659	6663	6676	6678
6699	6711	6716	6727	6744	6750	6752	6755	6762	6764
6778	6783	6797	6802	6814	6820	6823	6830	6841	6842
6848	6857	6858	6863	6872	6898	6906	6907	6911	6912
6921	6941	6943	6949	6951	6952	6978	7030	7043	7052
7055	7075	7081	7101	7137	7145	7148	7159	7172	7173
7175	7176	7201	7215	7219	7228	7262	7267	7307	7312
7321	7333	7338	7340	7362	7367	7375	7379	7399	7404

Table 1: Performance matrix of the LLM-hybrid system in a random 10% subset of the total screening pool (k=1704).
Green = Exclusion consensus, Blue = Inclusion consensus, Yellow = Incorrectly included by LLM, Red = Incorrectly excluded by LLM.

	Human - Inclusion	Human - Exclusion
LLM - Inclusion	6	0
LLM - Exclusion	0	164

100% agreement between the LLM and manual human screening.

Sensitivity: 100%, Specificity: 100%

Cohen's Kappa (K) = 1.00 (Perfect agreement)

Notes:

- **Duplicate management (ID 5991, 6001):**
Trials are duplicates. Original articles were assessed and correctly included.
- **Data quality (ID 5752):**
Both the LLM and human reviewers included this study during initial screening; however, it was subsequently excluded during full-text review.

Phase 2: Full-text screening (Trial IDs: 7419-7753)

7419	7437	7443	7455	7485	7497	7530	7543	7545	7548
7551	7558	7565	7566	7571	7577	7579	7585	7596	7614
7616	7617	7621	7623	7631	7662	7681	7695	7704	7711
7721	7729	7733	7738						

Table 2: Performance matrix of the LLM-hybrid system in a random 10% subset of the screening pool after exclusion of irrelevant studies (k=335). Green = Exclusion consensus, Blue = Inclusion consensus, Yellow = Incorrectly included by LLM, Red = Incorrectly excluded by LLM.

	Human - Inclusion	Human - Exclusion
LLM - Inclusion	7	1
LLM - Exclusion	0	26

97.1% agreement between the LLM and manual human screening.

Sensitivity: 100%, Specificity: 96.3%

Cohen's Kappa (K) = 0.915 [95%CI: 0.750 to 1.000] (Near perfect agreement)

Notes:

- **Methodological ambiguity (ID 7565):**
Human reviewers excluded the study due to unclear blood pressure measurement descriptions. The LLM initially included the study but correctly reverted to exclusion upon specific inquiry regarding the measurement protocol.
- **Inconsistent response coherence (ID 7561):**
The LLM initially flagged the study for inclusion but self-corrected to exclusion within the same response due to missing methodological details. This highlights the need for human oversight.
- **External verification and prompt compliance (ID 7662):**
The original entry is registered as a terminated trial, however, the LLM correctly identified the study for inclusion by independently retrieving information on the associated published trial via NCT number. While the correct conclusion was reached, it did not follow the prompt's prohibition against external web searching.
- **Duplicate management (ID 7704, 7733):**
Trials are duplicates. Original articles were assessed and correctly included.

Phase 3: Final inclusion (Trial IDs: 7754-7791)

7754	7755	7756	7757	7758	7759	7760	7761	7762	7763
7764	7765	7766	7767	7768	7769	7770	7771	7772	7773
7774	7775	7776	7777	7778	7779	7780	7781	7782	7783
7784	7785	7786	7787	7788	7789	7790	7791		

Table 3: Performance matrix of the LLM-hybrid system for all included trials in the primary analysis. Green = Exclusion consensus, Blue = Inclusion consensus, Yellow = Incorrectly included by LLM, Red = Incorrectly excluded by LLM.

	Human - Inclusion	Human - Exclusion
LLM - Inclusion	37	0
LLM - Exclusion	1	0

97.4% agreement between the LLM and manual human screening.

Sensitivity: 97.4%

Notes:

- **The limitations of LLMs (ID 7764):**

The LLM initially excluded this trial because the primary manuscript did not detail the blood pressure measurement methodology. Correct inclusion was only achieved after providing the model with 1) The main manuscript and its protocol 2) The pre-specified blood pressure sub-analysis article and 3) The specific Japanese Society of Hypertension (JSH) guidelines referenced in the sub-analysis. We attempted to allow the LLM to disregard the «no web searching» limitation, and it still failed to include the trials, unless manually provided all relevant documents.

- **Methodological insight:**

This case emphasizes that the model is strictly limited to the provided files and text and does not autonomously search for referenced protocols unless explicitly instructed. While one could argue this false exclusion could have been avoided if the original authors had been clearer in their main manuscript, it reinforces the necessity of a "Human-in-the-loop" hybrid approach to handle trials relying on external methodological standards, ensuring that clinical nuance and decentralized protocols are not lost in automated screening pipelines.